MILITARY SPECIFICATION

PRACTICE WARHEAD SECTION: M65E3

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of practice warhead section entitled Practice Warhead Section: M65E3 which is used for training on the Sergeant warhead.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure For.

DRAWINGS

ORDNANCE CORPS

8821748	- Barosetter.
8838222	- Adapter Ring.
8838468	- Safe Plug.
8838469	- Safe Plug.
8838624	- Vernier Weight.
8850565	- Shipping and Storage Container M481. Sgt. Adaption Kits and Warhead Sections, Packing and Marking.
8838578	- Ballast, Weight.
8857686	- Practice Warhead Section M65E3.
8857848	- Ballast Rear.
8858606	- Static Tube, Dummy, M16.
EL-8854177	- Equipment List.

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The practice warhead section shall comply with all requirements specified on the drawing (dwg) 8857686, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. The contractor shall assign serial numbers as required by drawings. The contractor may select any number which complies with the drawing requirement. These numbers shall not be duplicated.

3.5 Weight. The Practice Warhead Section M65E3 with Static Tube M16 attached shall weigh 1611 ± 20 pounds (lbs) when determined as specified in 4.4.1.

3.6 Insulation resistance. The insulation resistance between the pins of connectors as designated in Table I shall be 20 megohms minimum when tested as specified in 4.4.2. The safe plug (see dwg 8838469) shall be installed for this test.

Table I

Connector P919

From Pin	To Pins
4	9, 13, 15, 24, 35, 36, 50, 62, Case
9	4, 13, 15, 24, 35, 36, 50, 62, Case
13	4, 9, 15, 24, 35, 36, 50, 62, Case
15	4, 9, 13, 24, 35, 36, 50, 62, Case
24	4, 9, 13, 15, 35, 36, 50, 62, Case
35	4, 9, 13, 15, 24, 36, 50, 62, Case

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36	4, 9, 13, 15, 24, 35, 50, 62, Case
50	4, 9, 13, 15, 24, 35, 36, 62, Case
62	4, 9, 13, 15, 24, 35, 36, 50, Case
Case	4, 9, 13, 15, 24, 35, 36, 50, 62

Connector J16

From Pin A	To Case

3.7 Circuit resistance and continuity. The resistance between connector pins shall be as designated when the stepping switch is in the positions indicated in Table II. Pulse the stepping switch with 28 volts direct current (vdc) plus or minus (±) 10 percent (%) with a pulse width of less than one second. The positive pulse shall be applied to P919-31 and the return shall be connected to P919-18. Test as specified in 4.4.3. The safe plug (see dwg 8838468) shall be inserted for this test.

Table II

Ledex Position	Pin to Pin		Resistance (Ohms)	Tolerance (± %)
	P919-9	P919-36	0.2 max	
	P919-9	P919-42	0.2 max	
	P919-9	P919-3	0.2 max	
	P919-21	P919-14	0.2 max	
	P919-21	P919-27	0.2 max	
	P919-21	P919-40	0.2 max	
	P919-21	P919-1	0.2 max	
	P919-21	P919-8	0.2 max	
	P919-24	P919-31	0.2 max	
	P919-50	P919-70	0.2 max	
	J16-A	J16-B	0.2 max	
	J16-A	J16-C	0.2 max	

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Ledex Position	Pin to Pin		Resistance (Ohms)	Tolerance (± %)
	P919-15	P919-22	715 **	10
	P919-18	P919-24	10 min **	
	P919-11	P919-4	290 **	10
	P919-11	P919-17	440 **	10
	P919-28	P919-35	30,000	10
	P919-62	P919-68	11,000	0.1
1 *	P919-50	P919-69	3,000	0.1
2 *	P919-50	P919-68	3,000	0.1
3 *	P919-50	P919-68	5,000	0.1
4 *	P919-50	P919-68	7,000	0.1
5 *	P919-50	P919-68	9,000	0.1
6 *	P919-50	P919-68	9,000	0.1

* Apply 28 vdc to pins P919-31 and P919-18 to pulse Ledex.
** Measure at 80° ± 20°F only.

3.8 Operation. Each practice warhead section shall operate and shall function as specified when tested as specified in 4.4.4.

3.9 Environmental testing (see 4.4.5).

3.9.1 Low temperature. The practice warhead section shall withstand exposure to a temperature of -65 ± 5 degrees Fahrenheit (°F) without damage and while at this temperature shall meet the requirements of 3.6, 3.7 and 3.8 when tested as specified in 4.4.5.1.

3.9.2 High temperature. The practice warhead section shall withstand exposure to a temperature of 160 ± 5°F without damage and while at this temperature shall meet the requirements of 3.6, 3.7, and 3.8 when tested as specified in 4.4.5.2.

3.9.3 Transportation vibration. The practice warhead section, packed in accordance with dwg 8850565 in the shipping and storage container, shall withstand vibration in an axis perpendicular to the container base surface at the frequencies and amplitudes specified in Table III without evidence of damage or affecting function when tested as specified in 4.4.5.3. Upon completion of the vibration test, the practice warhead section shall meet the requirements of 3.6, 3.7, and 3.8.

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Table III

Frequency Range Hertz (Hz)	Constant Acceleration or Double Amplitude	Cycle Time
6-20-6	1 g ± 0.1 Gravity (g)	15 min
21-60-21	0.060 ± 0.006 inch	10 min
61-300-61	10 g ± 1 g	25 min
301-500-61	10 g ± 1 g	25 min

3.9.4 Shock. The practice warhead section, packed in the shipping and storage container in accordance with dwg 8850565, shall withstand shocks of edgewise drop, end impact and rollover impact without evidence of damage or impairment of operation when tested as specified in 4.4.5.4. Upon completion of the shock tests, the practice warhead section shall meet the requirements of 3.6, 3.7, and 3.8.

3.10 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460(MU).

3.11 Workmanship.

3.11.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.11.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.11.3 Burrs. Parts shall be free from burrs.

3.11.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.11.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.11.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 1 complete assembly (see dwg 8857686) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table IV. Tests shall be conducted in order specified in Table IV.

Table IV

First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Weight (see 3.5)		X
3. Insulation resistance (see 3.6)		X
4. Circuit resistance and continuity (see 3.7)		X
5. Operation (see 3.8)		X
6. Low temperature (see 3.9.1)		X
7. High temperature (see 3.9.2)		X
8. Transportation vibration (see 3.9.3)		X
9. Shock (see 3.9.4)	X	

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4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table V and Table VI), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Table V
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

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Table VI
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Adapter ring (dwg 8838222).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major: None defined.

Minor:

201. Location of alignment pin holes.....Gage
202. Diameter, alignment pin holes.....Gage
203. Location of mounting holes.....Gage
204. Inside diameter.....Gage
205. Incorrect or illegible marking.....Visual
206. Bare spot in protective coating except for slight scratches.....Visual
207. Evidence of poor workmanship (see 3.11).....Visual

4.3.2.1.2 Ballast, weight (dwg 8857578).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:

101. Mislocation of vernier weight mounting holes.....Gage
102. Incorrect concentricity of forward diameter.....Gage
103. Mislocation of handling equipment mounting holes.....Gage

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104. Handling equipment threaded mounting holes oversize.....Gage

Minor:
201. Scratches in protective finish with base metal exposed.....Visual
202. Evidence of poor workmanship (see 3.11).....Visual

4.3.2.1.3 Vernier weight (dwg 8838624).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:
101. Mislocation of mounting holes.....Gage

Minor:
201. Scratches in protective finish with base metal exposed.....Visual

4.3.2.1.4 Ballast rear (dwg 8857848).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:
101. Location of mounting holes.....Gage
102. Location of handling equipment holes.....Gage

Minor:
201. Bare spot in protective coating except for slight scratches.....Visual

4.3.2.1.5 Static tube, dummy, M16 (dwg 8858606).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major: None defined.

Minor:
201. Scratches in protective finish with base metal exposed.....Visual
202. Incorrect or improper marking.....Visual

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4.3.2.1.6 Practice warhead section: M65E3 (dwg 8857686).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Mismatch at station 8.125 and 54.75 exceeds 0.020	Gage
102.	Gap in skin contour at station 8.125, 54.75, and 139.0 exceeds 0.020	Gage
103.	Overall length tolerance of assembly from station 8.125 to 139.0 exceeds + 0.312 inches	Gage
104.	Incorrect weight	Gage
105.	Incorrect torque on bolts	Gage
106.	Incorrect mating of connectors	Visual
107.	Any component missing, loose or damaged	Visual
108.	Protruding fasteners	Visual
109.	Objective evidence relative to material missing or improper (see 3.1)	Visual
110.	Serialization improper or missing (see 3.4)	Visual
Minor:		
201.	Bare spot in protective coating except for slight scratches	Visual
202.	Improper or illegible markings	Visual

4.3.2.1.7 Packaging for practice warhead section: M65E3 (dwg 8850565) prior to inserting warhead section.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Loose internal component	Visual/Manual
102.	Damaged internal component	Visual
Minor:		
201.	Foreign matter in container	Visual

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4.3.2.1.8 Packing for practice warhead section: M65E3 (dwg 8850565) prior to closing container.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Incorrect torque of bolts	Gage
102.	Insecure fastening of static probe package	Visual/Manual
103.	Incorrect tension of straps	Visual/Manual
Minor:		
201.	Missing dust caps	Visual
202.	Loose dust caps	Visual/Manual

4.3.2.1.9 Packing for practice warhead section: M65E3 (dwg 8850565).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Loose latches	Manual
102.	Insufficient desiccant	Visual
103.	Missing unexposed humidity indicator	Visual
104.	Incorrect location of center of gravity band	Visual
Minor:		
201.	Missing records	Visual
202.	Missing car seals	Visual
203.	Incorrect or illegible marking	Visual
204.	Damaged component	Visual
205.	Bare spot in protective coating except for slight scratches	Visual

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4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test Classification
a. Weight	Major
b. Insulation resistance	Major
c. Circuit resistance and continuity	Major
d. Operation	Major

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Weight, insulation resistance, circuit resistance and continuity, and operation. Each practice warhead section shall be tested in accordance with 4.4.1, 4.4.2, 4.4.3, and 4.4.4. If any item fails to comply with 3.5, 3.6, 3.7, and 3.8, the item shall be rejected and removed from the lot.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8854177). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Weight. The total weight of the M65 practice warhead section shall be determined by weighing on a suitable scale.

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4.4.2 Insulation resistance. Insulation resistance shall be measured by means of an insulation resistance tester at 500 + 50 vdc. The maximum current output of the insulation resistance tester shall not exceed 5 milliamperes. All selector switches shall be in Position 1 for this test unless otherwise specified (see Table I).

4.4.3 Circuit resistance and continuity. A suitable electrical resistance measuring instrument shall be connected to the specified connector pins for the measurement of circuit resistance (see Table II).

4.4.4 Operation. Step the stepping switch through at least one complete revolution. The switch shall be stepped and monitored as indicated in Table II.

4.4.4.1 Setting. Connect a 680 + 5% ohm, 5 watt resistor to pins P919-8 and P919-11. Connect a Bendix Barosetter Type 8821748 to the appropriate pins in terminal P919 as shown in Figure 1. Set the barosetter at -1300 feet. Press the setting button until a null is indicated (a null is indicated on the Bendix Barosetter when the green "null" light comes on). Repeat the above for settings of 10,000 and 25,000 feet. Measure the voltage between pins P919-8(+) and P919-11(-). This voltage shall be 46 + 6 vdc as long as the barosetter is connected and the power is applied. CAUTION: BE SURE THE 680 OHM RESISTOR IS CONNECTED TO PINS P919-8 AND P919-11 BEFORE ANY POWER IS APPLIED.

4.4.4.2 Diode line. Disconnect the barosetter. Connect the positive side 200 + 10% vdc to pin P919-8. Connect the negative side of the 200 vdc in series with a 100 + 5% kilo-ohm (K), 0.5 watt resistor to pin J14-A. Measure the voltage across the 100 K resistor. The measured voltage across the 100 K resistor shall not exceed 1.0 volt.

4.4.5 Environmental testing (see 3.9).

4.4.5.1 Low temperature. The practice warhead section shall be placed in a test chamber. The chamber's temperature shall be lowered and stabilized at -65 + 5°F (see 6.4). After the practice warhead section has been subjected to this stabilized temperature for a minimum of 4 hours and while still at this temperature, it shall be tested for compliance with the requirements of 3.6, 3.7, and 3.8.

4.4.5.2 High temperature. The practice warhead section shall be placed in a test chamber. The chamber's temperature shall be raised and stabilized at 160 + 5°F (see 6.4). After the practice warhead section has been subjected to this stabilized temperature for a minimum of 4 hours and while still at this temperature, it shall be tested for compliance with the requirements of 3.6, 3.7, and 3.8.

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4.4.5.3 Transportation vibration. Place the shipping and storage container containing the practice warhead section on a suitable vibration machine and vibrate for the durations, frequencies and amplitudes specified in Table III. At the completion of this test, the warhead section shall be inspected for evidence of damage and tested for the requirements of 3.6, 3.7, and 3.8.

4.4.5.4 Shock. The shipping and storage container containing the practice warhead section shall be subjected to the shocks specified in 4.4.5.4.1 through 4.4.5.4.3. Upon completion of these shocks, the warhead section shall be inspected for evidence of damage and tested for compliance with the requirements of 3.6, 3.7, and 3.8.

4.4.5.4.1 Edgewise drop. Support the loaded shipping and storage container on one end of its base on a wood sill 5 ± 0.5 inches high and placed normal to the skids. The opposite end of the container shall be raised and allowed to drop freely from a height of 12, 24, and 36 ± 0.5 inches respectively. The three drops shall be made on concrete or a similar hard surface. Apply the test to both ends of the container.

4.4.5.4.2 End impact. The loaded shipping and storage container shall be placed in a fixture which will allow free axial movement. The end of the container shall strike against a solid flat vertical abutment having a wooden face of 2 inch minimum thickness. The direction of motion shall be 90 ± 5° with the impact surface and the velocity at impact shall be 10.5 ± 1.5 feet per second. Apply the test to both ends of the container.

4.4.5.4.3 Rollover impact. The loaded shipping and storage container on its base on a hard level floor shall be tipped slowly sideways until it falls freely on its side. The procedure shall be repeated from the side to the top, from the top to the other side, and from the other side to its base, thus completing one revolution.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The practice warhead section shall be packaged in accordance with dwg 8850565 using the specified moisture barrier.

5.1.2 Level C. The practice warhead section shall be packaged in accordance with dwg 8850565 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The practice warhead section shall be packed in accordance with dwg 8850565.

5.3 Marking. Marking shall be as specified in accordance with dwg 8850565.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-2262 (Rev 1) dated 29 October 1965.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1115-A054

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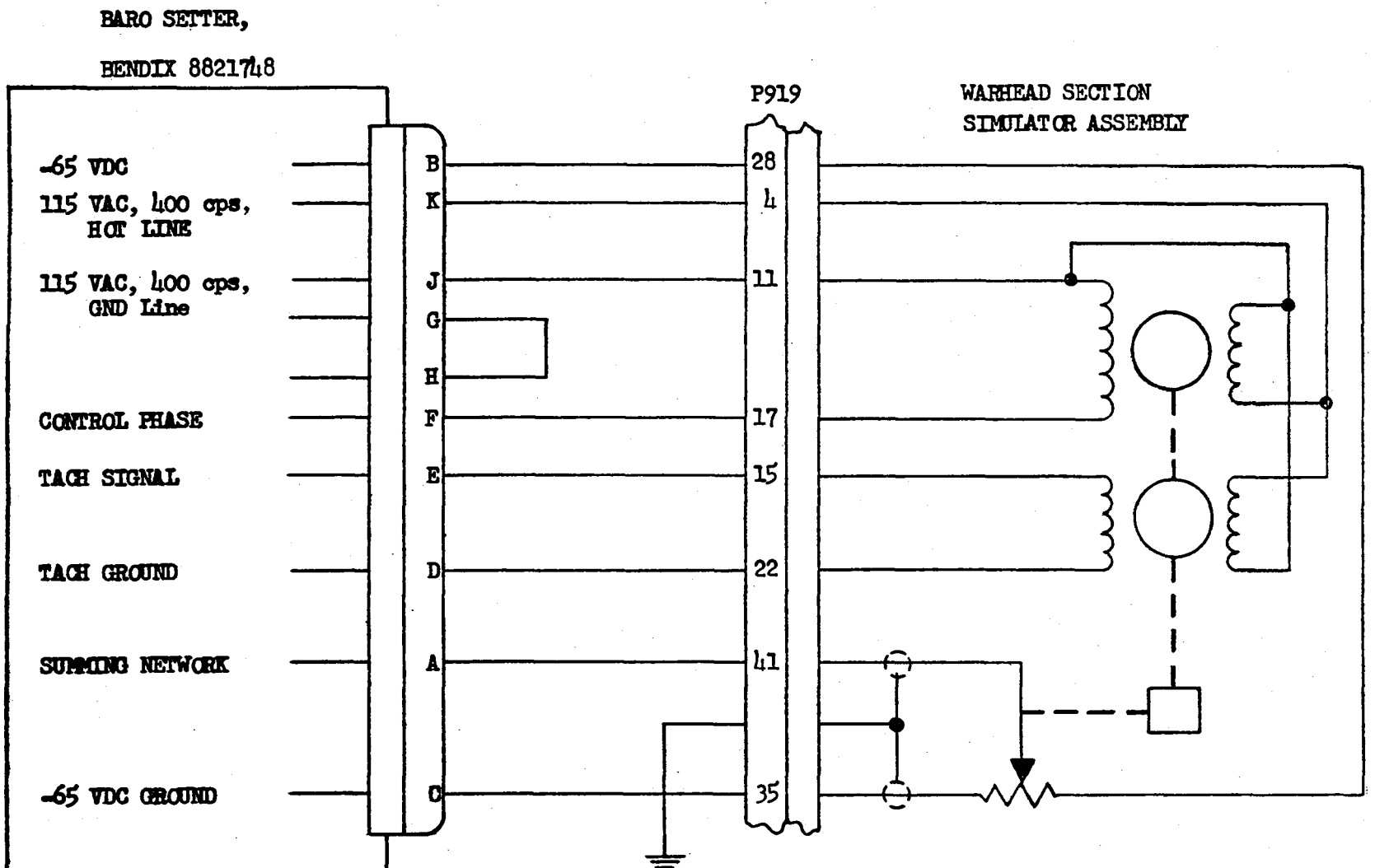


Figure 1. Test Circuit

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